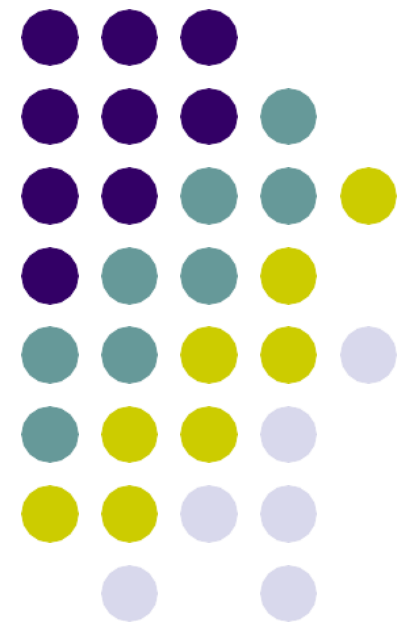


Biomedical Image Processing

*Biomedical Engineering Department
Institut Teknologi Sepuluh Nopember*





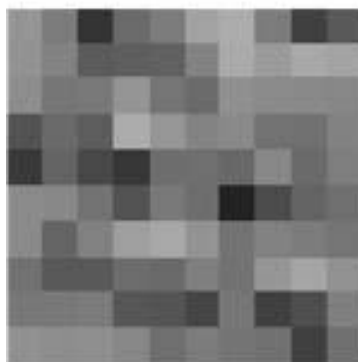
What is an Image?

- 2-dimensional matrix of Intensity (gray or color) values

Set of Intensity values

Image coordinates
are integers

$$I(u, v) \in \mathbb{P} \quad \text{and} \quad u, v \in \mathbb{N}.$$



$F(x, y)$

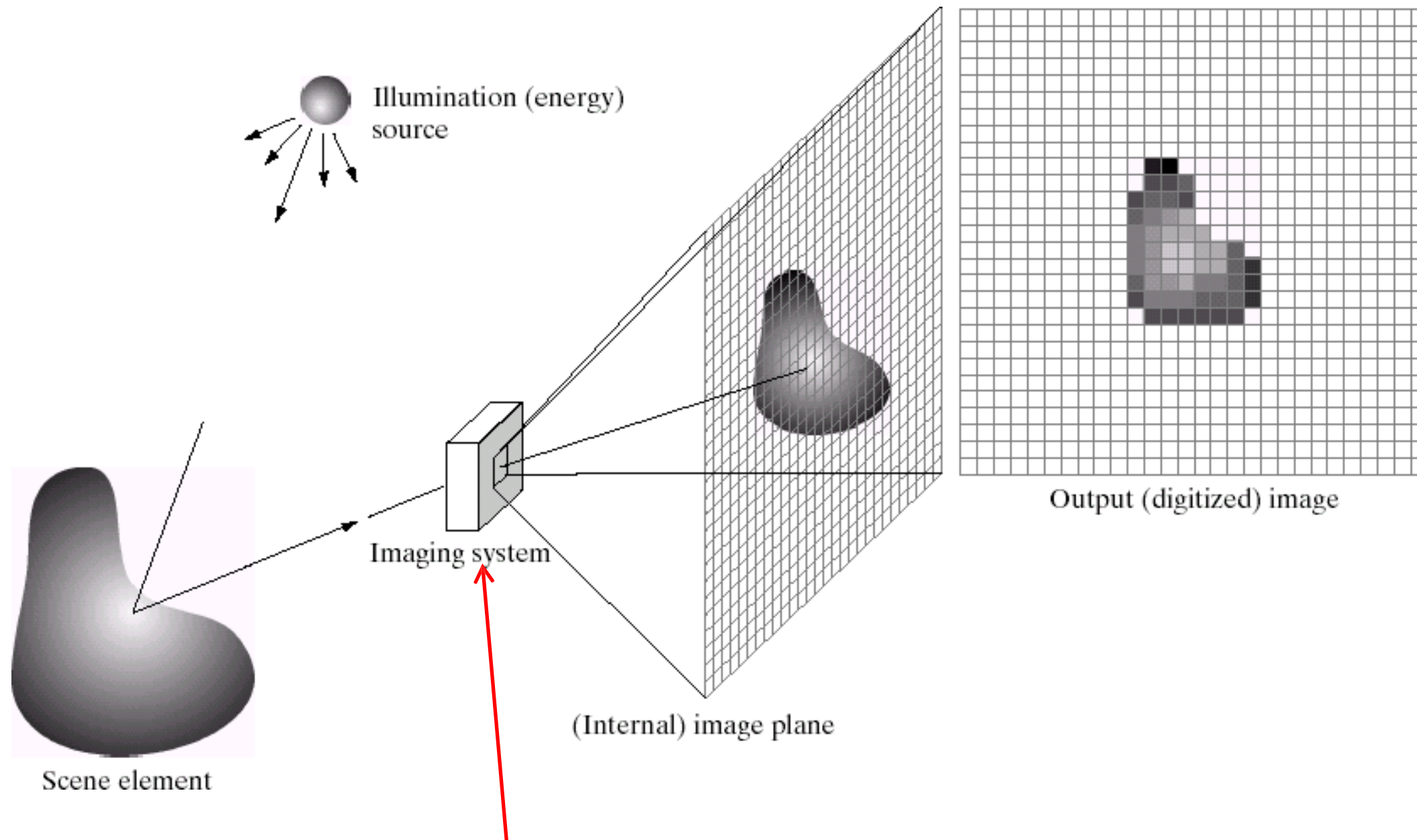


148	123	52	107	123	162	172	123	64	89	...
147	130	92	95	98	130	171	155	169	163	...
141	118	121	148	117	107	144	137	136	134	...
82	106	93	172	149	131	138	114	113	129	...
57	101	72	54	109	111	104	135	106	125	...
138	135	114	82	121	110	34	76	101	111	...
138	102	128	159	168	147	116	129	124	117	...
113	89	89	109	106	126	114	150	164	145	...
120	121	123	87	85	70	119	64	79	127	...
145	141	143	134	111	124	117	113	64	112	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

$I(u, v)$



Imaging System (Camera)



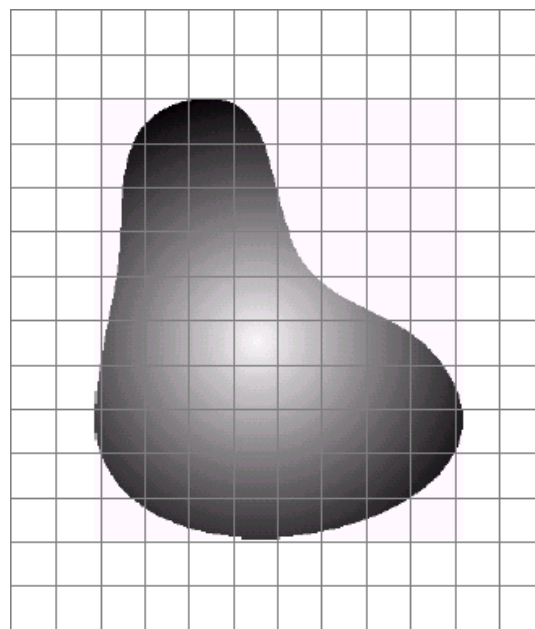
Example: a camera
Converts light to image

Credits: Gonzales and Woods

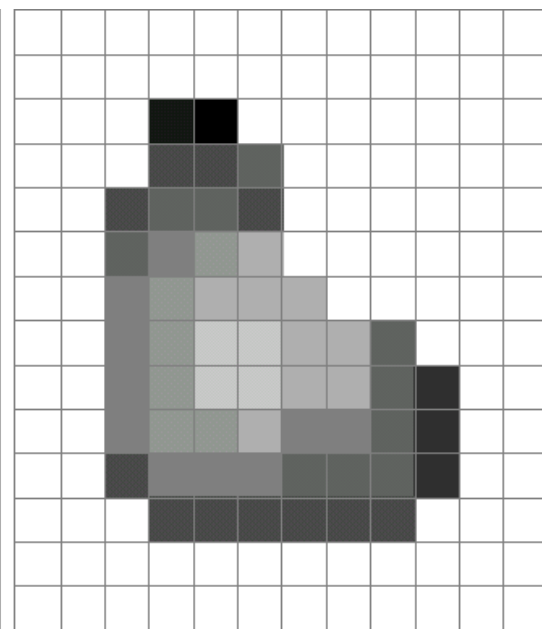


Digital Image?

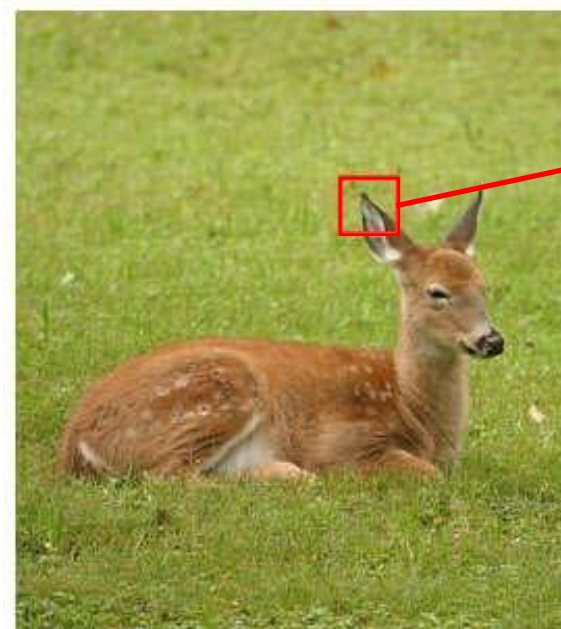
- Remember: *digitization* causes a digital image to become an *approximation* of a real scene



Real image



Digital Image
(an approximation)



Real image



Digital Image
(an approximation)



Digital Image

- Common image formats include:
 - 1 values per point/pixel (B&W or Grayscale)
 - 3 values per point/pixel (Red, Green, and Blue)
 - 4 values per point/pixel (Red, Green, Blue, + “Alpha” or Opacity)



Grayscale



RGB



RGBA

- We will start with gray-scale images, extend to color later



What is image Processing?

- Algorithms that alter an input image to create new image
- Input is image, output is image



Original Image



Processed Image

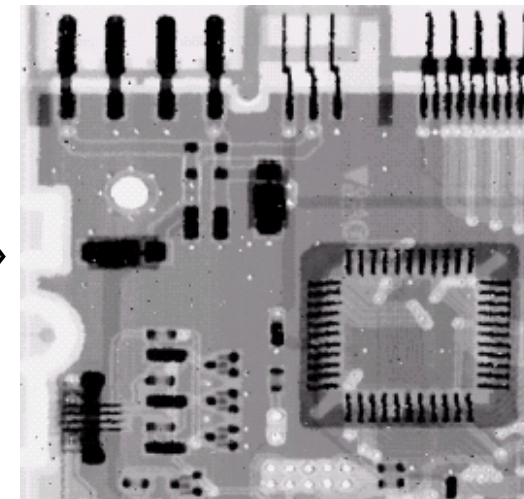
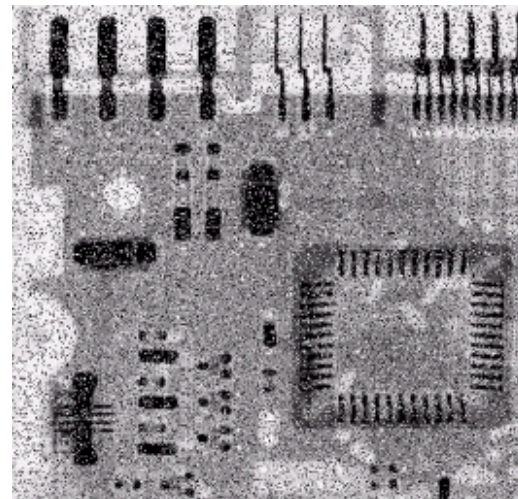
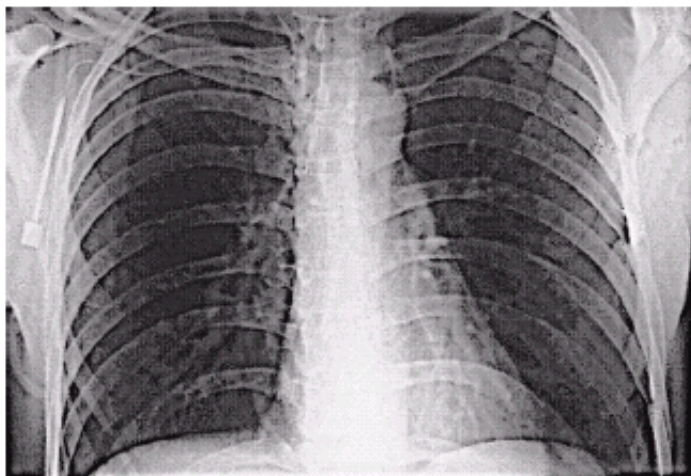
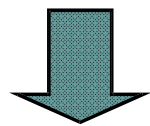
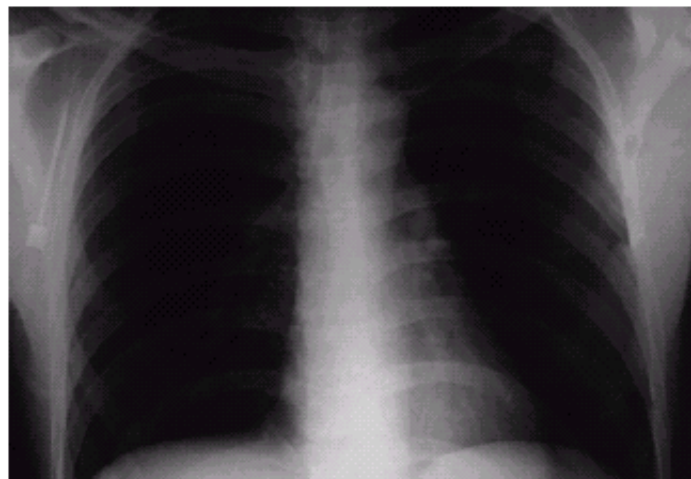
- Improves an image for human interpretation in ways including:
 - Image display and printing
 - Image editing
 - Image enhancement
 - Image compression

Example Operation: Noise Removal

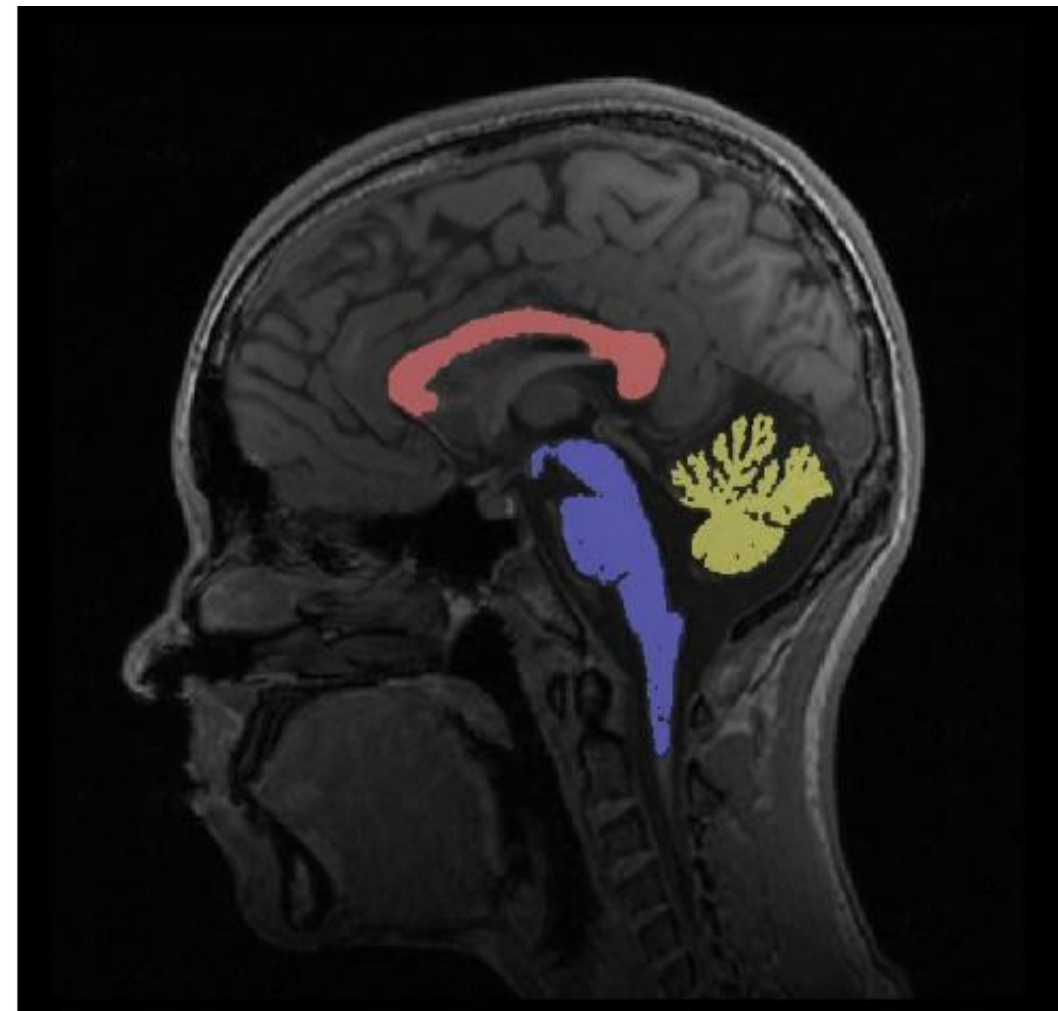
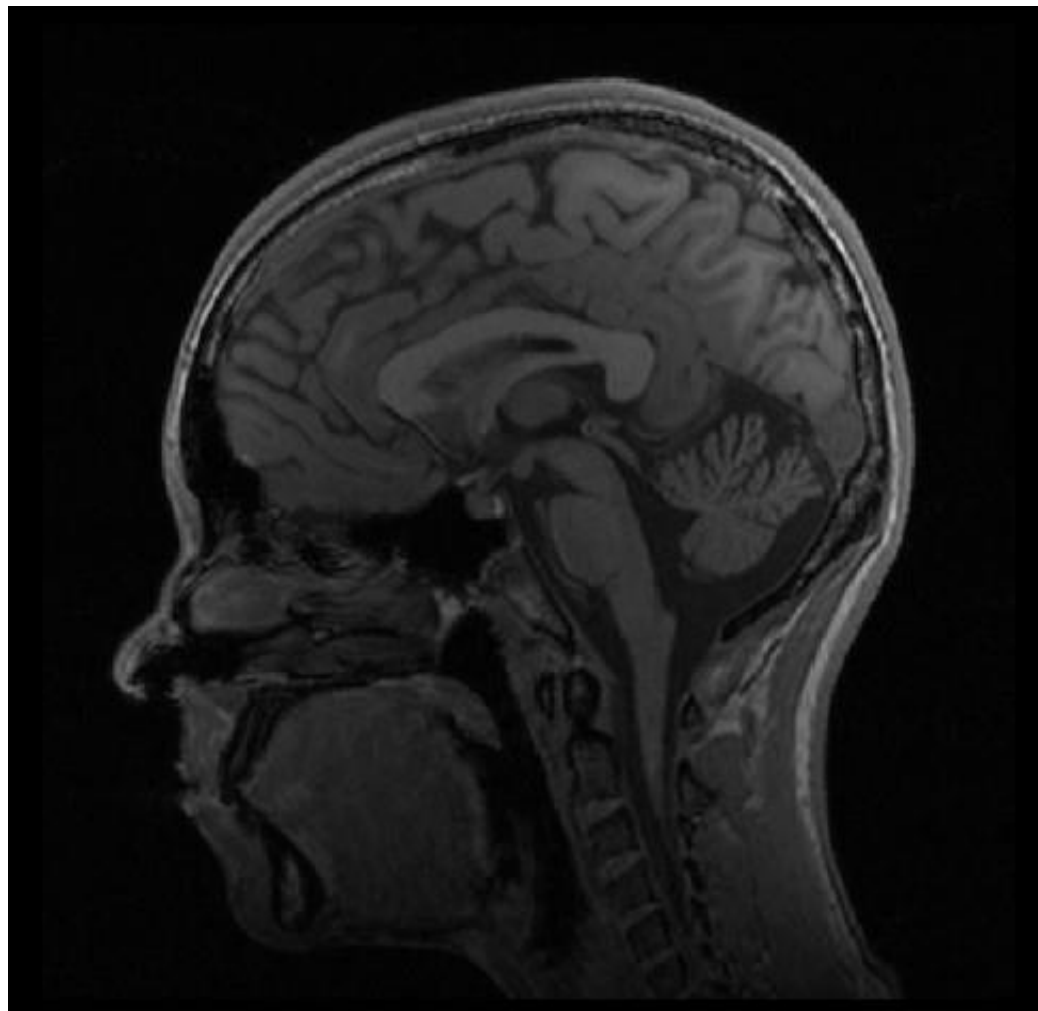


Noisy Image

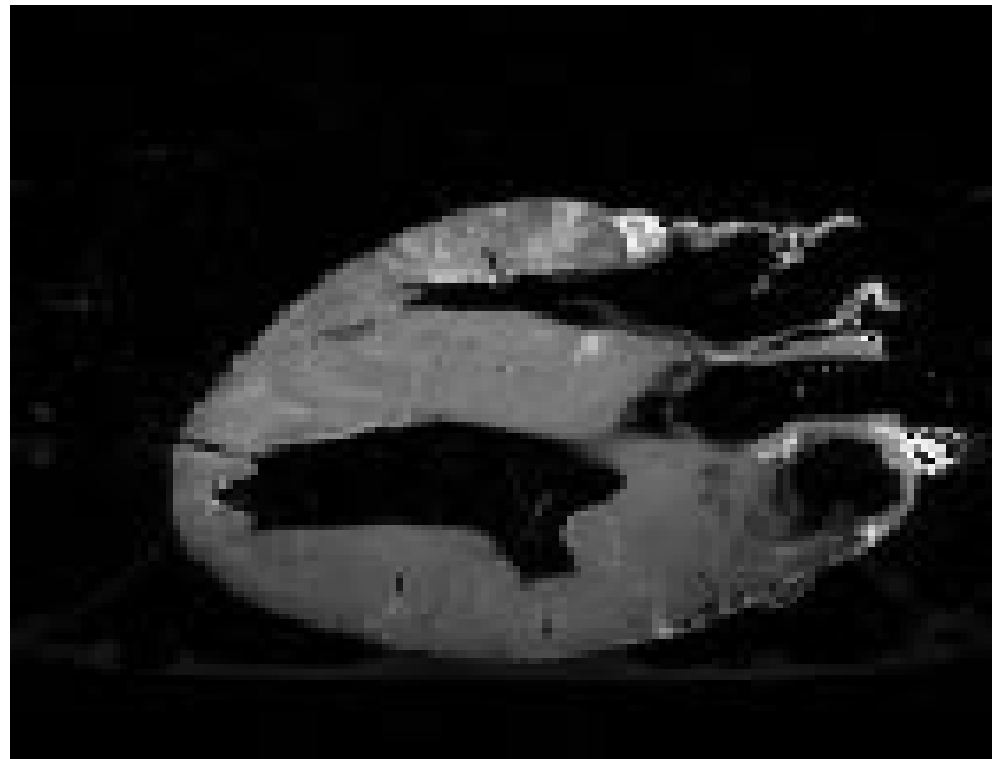
Denoised Image



Example: Region Detection, Segmentation



Applications of Image Processing: Medicine

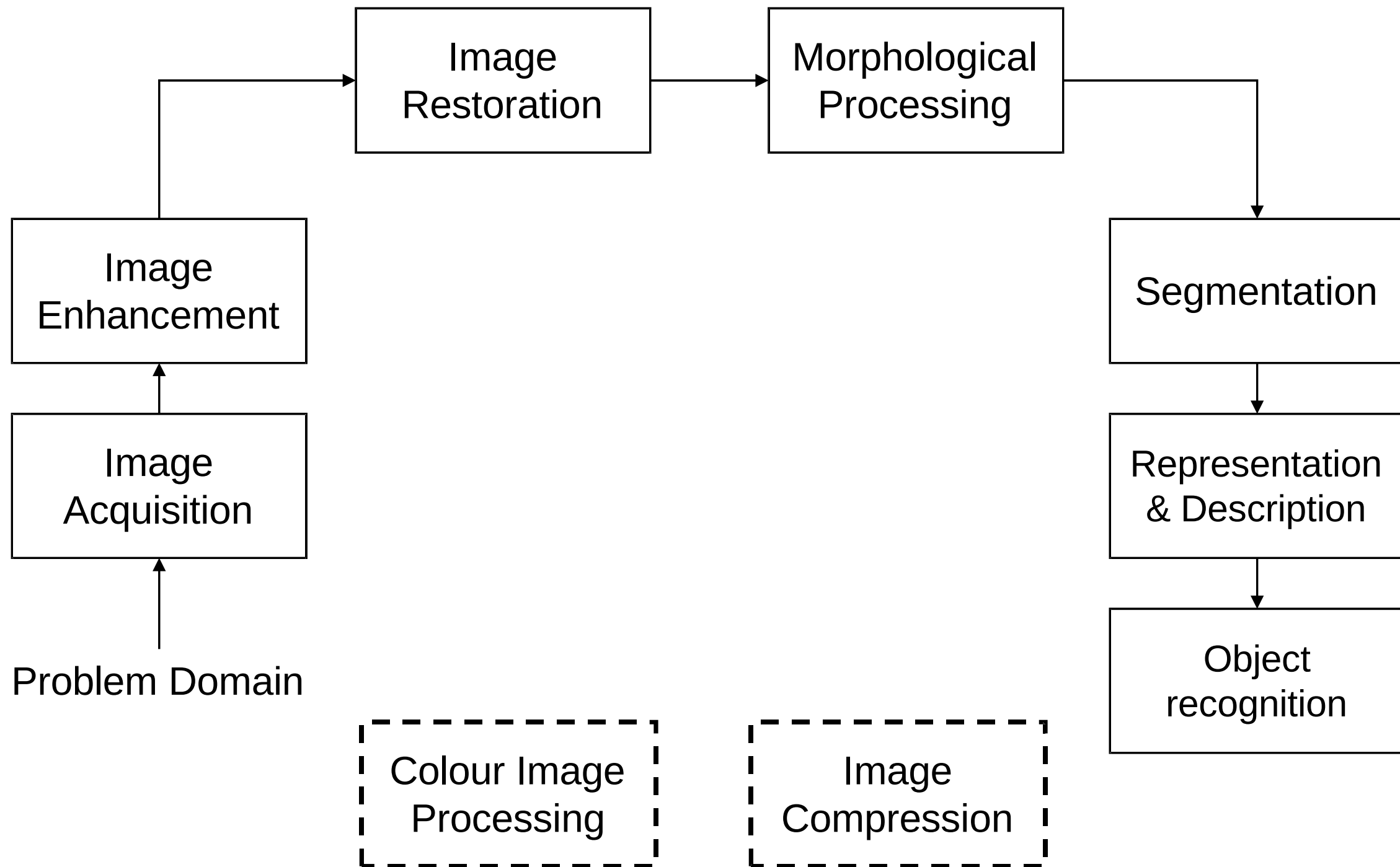


Original MRI Image of a Dog Heart

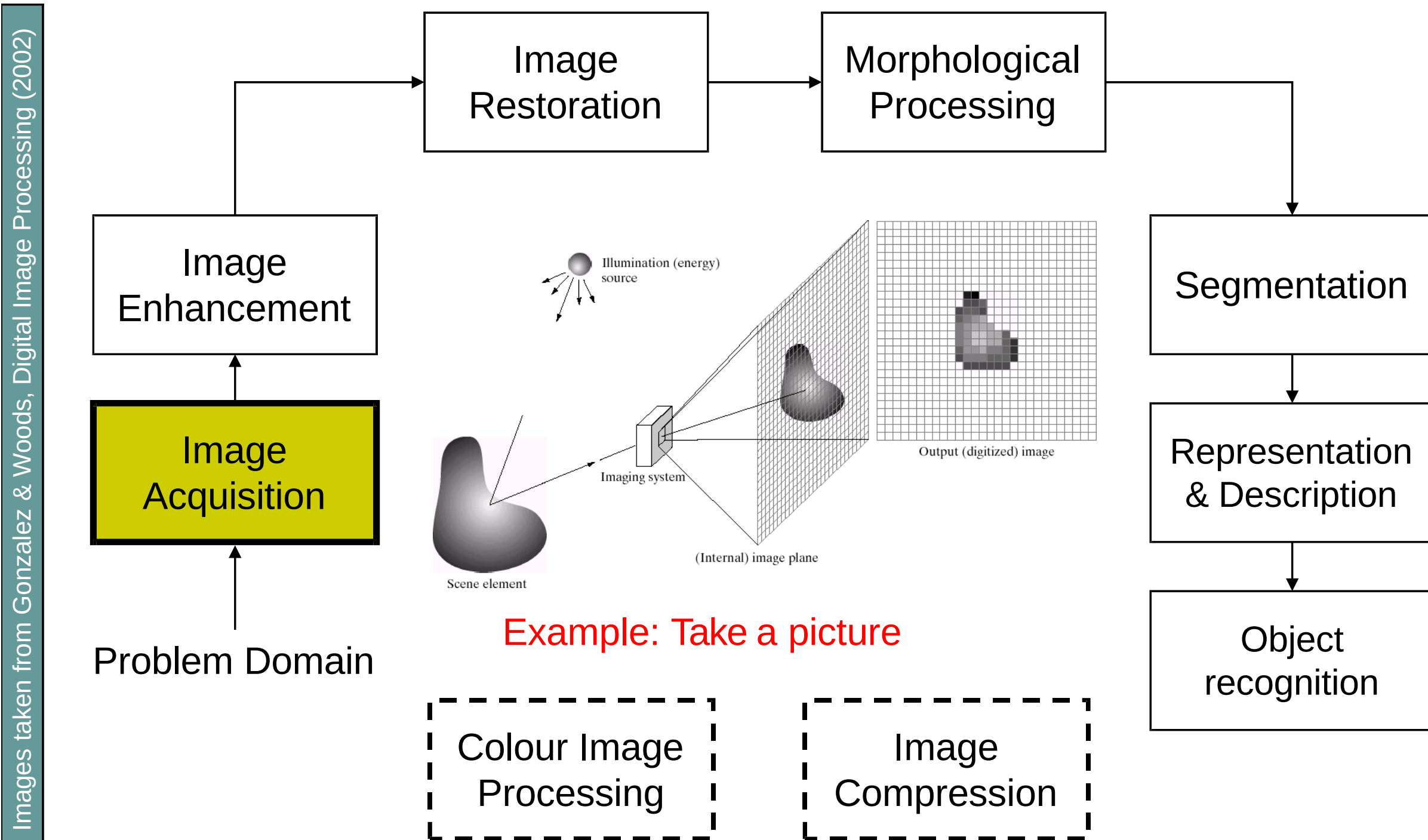


Edge Detection Image

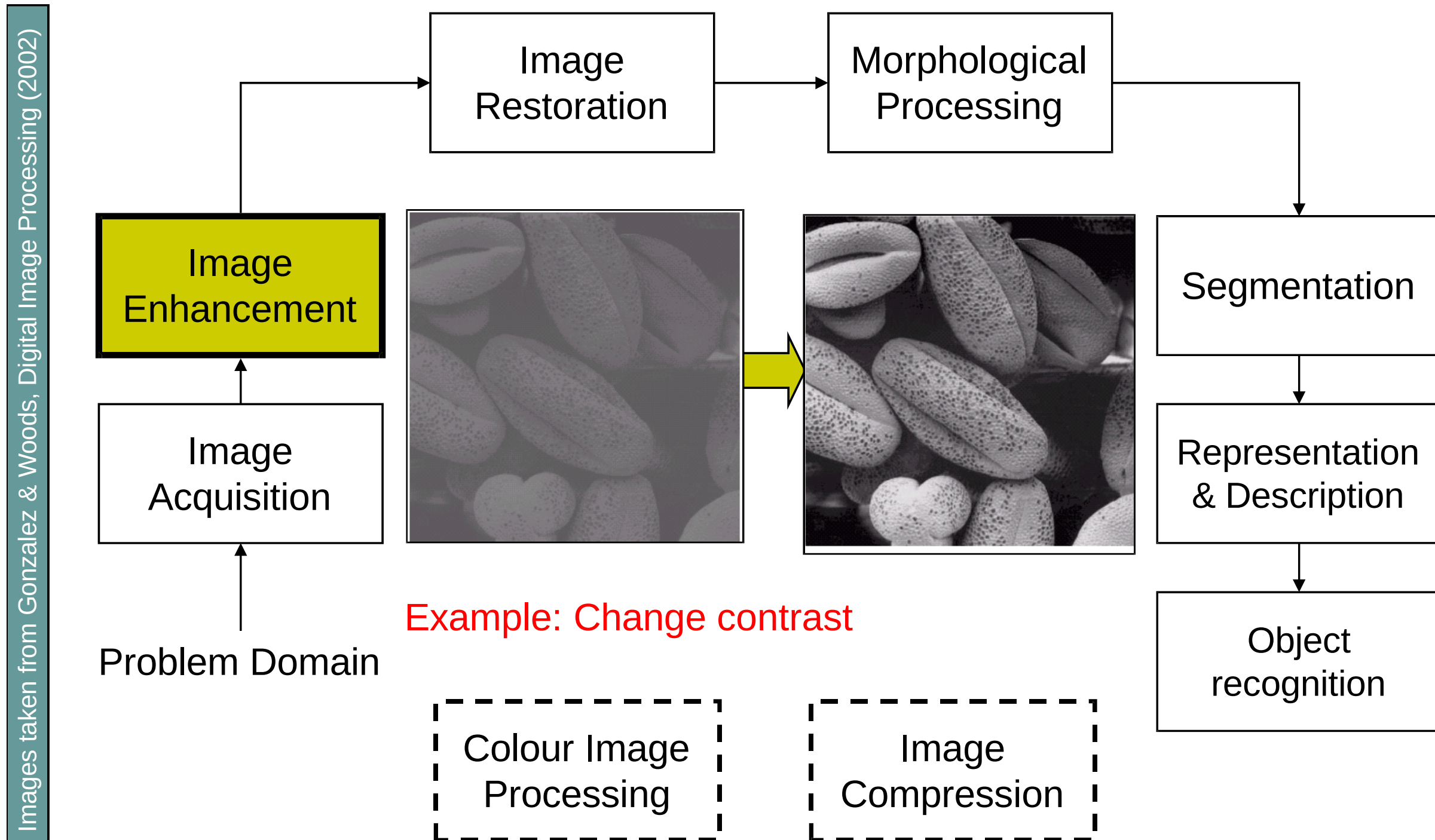
Key Stages in Digital Image Processing



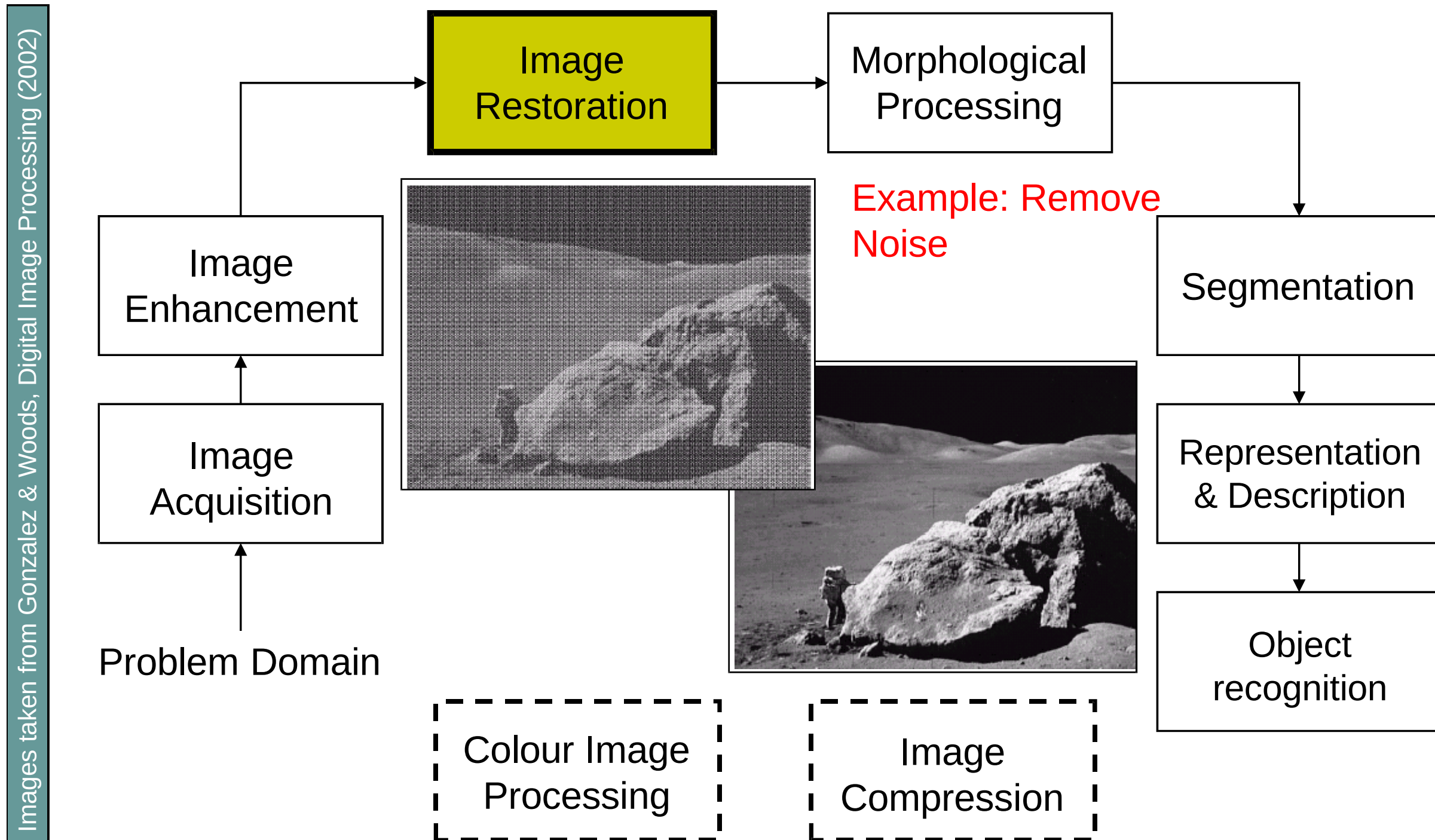
Key Stages in Digital Image Processing: Image Aquisition



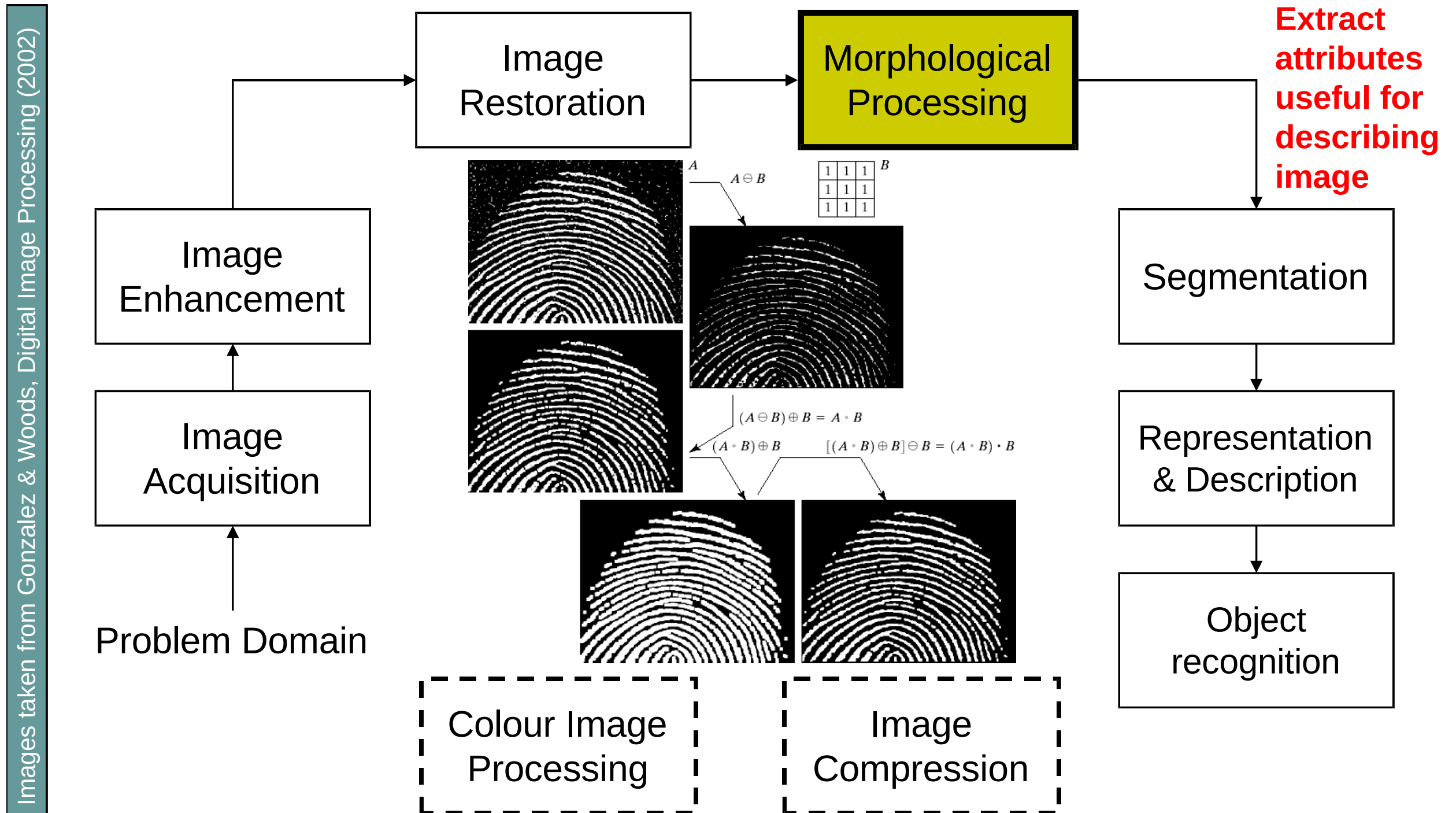
Key Stages in Digital Image Processing: Image Enhancement



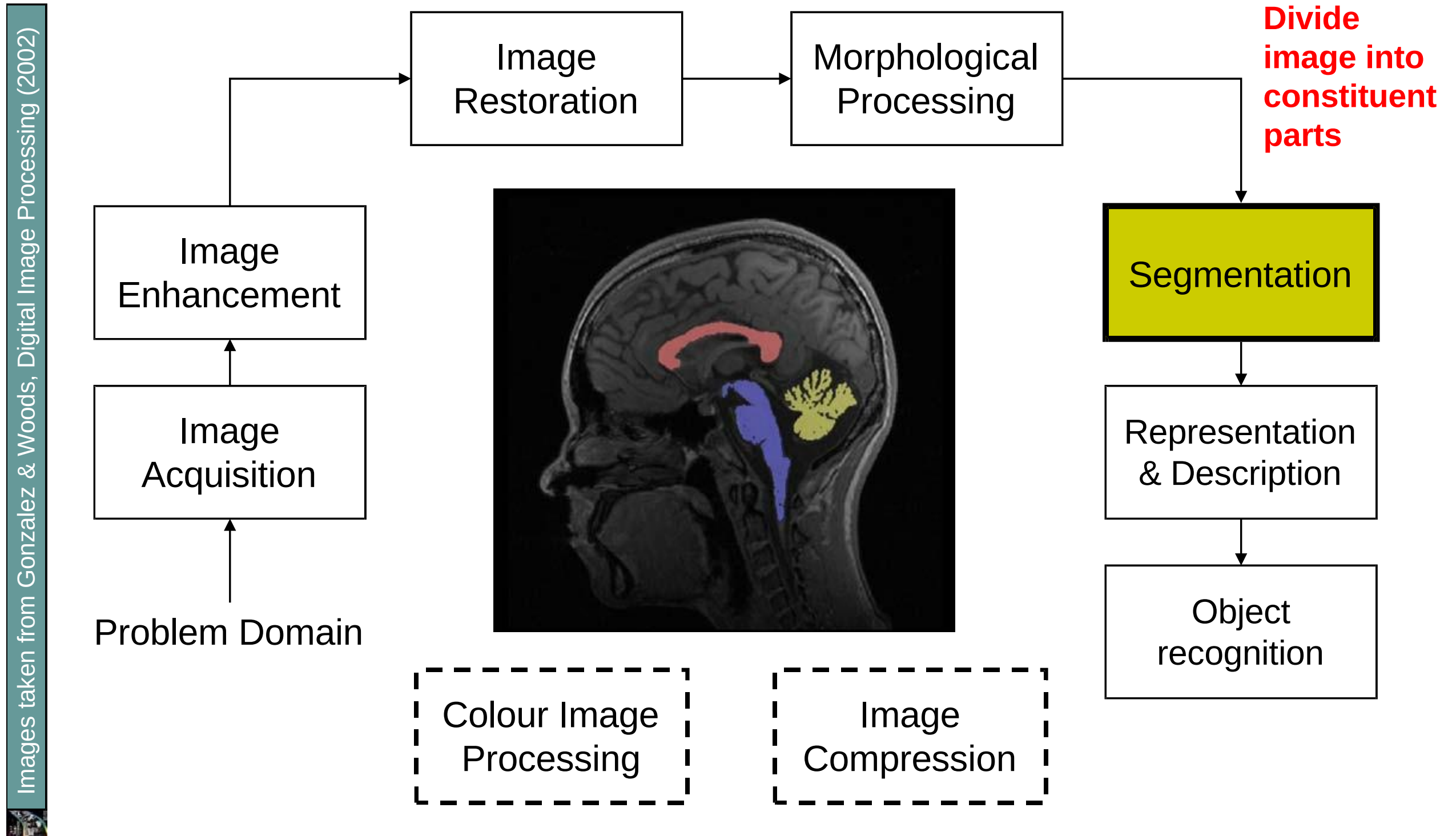
Key Stages in Digital Image Processing: Image Restoration



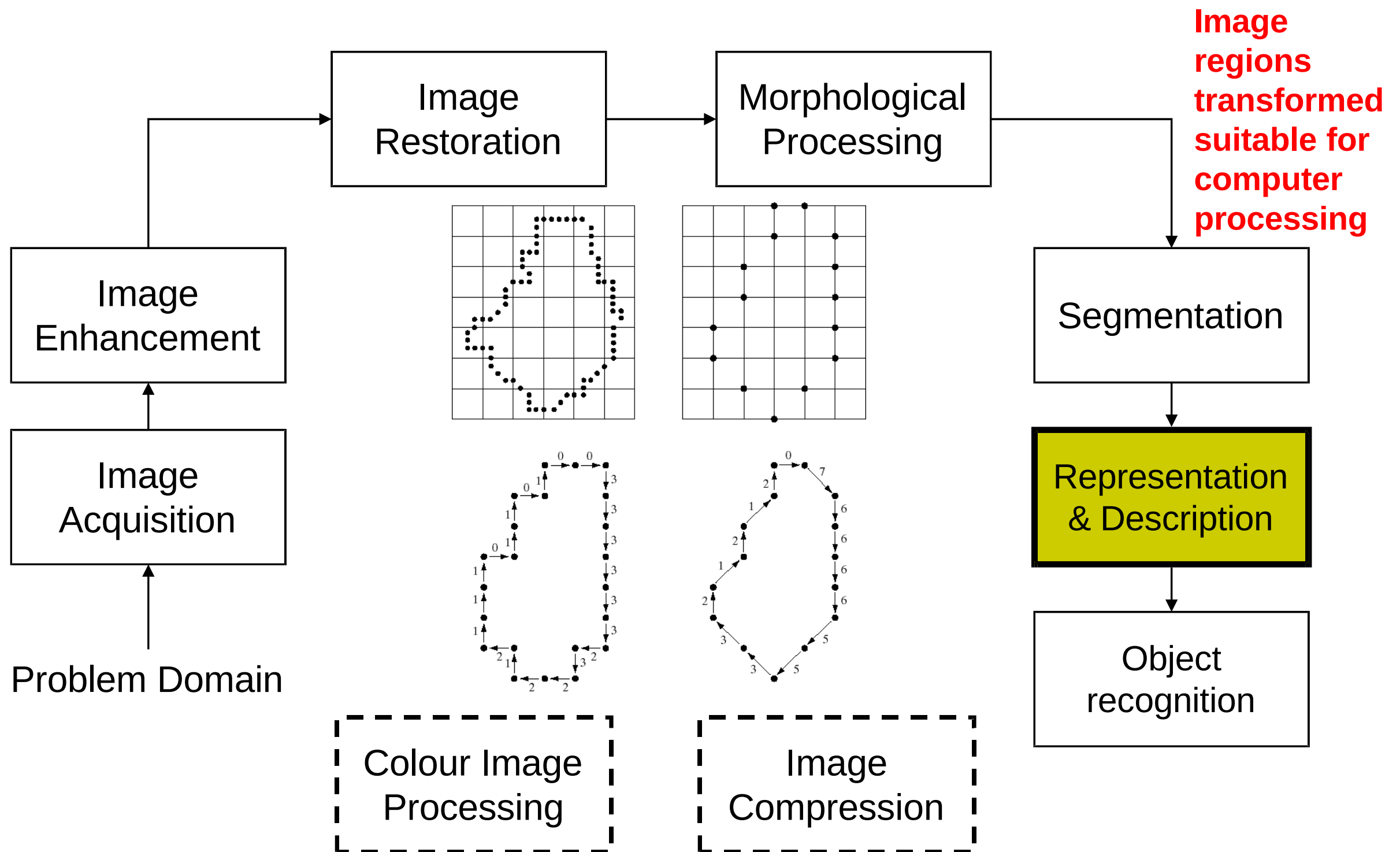
Key Stages in Digital Image Processing: Morphological Processing



Key Stages in Digital Image Processing: Segmentation



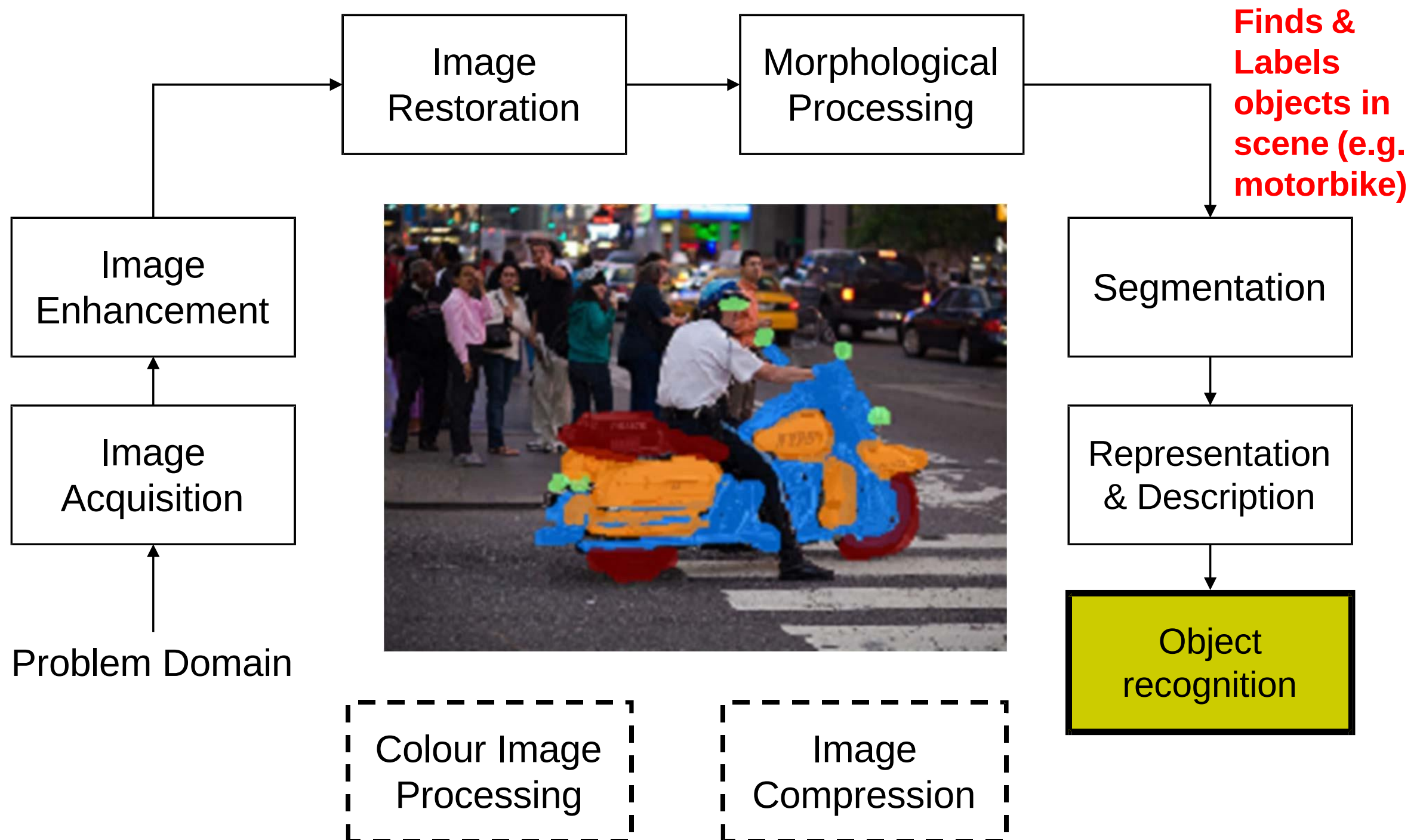
Key Stages in Digital Image Processing: Representation & Description



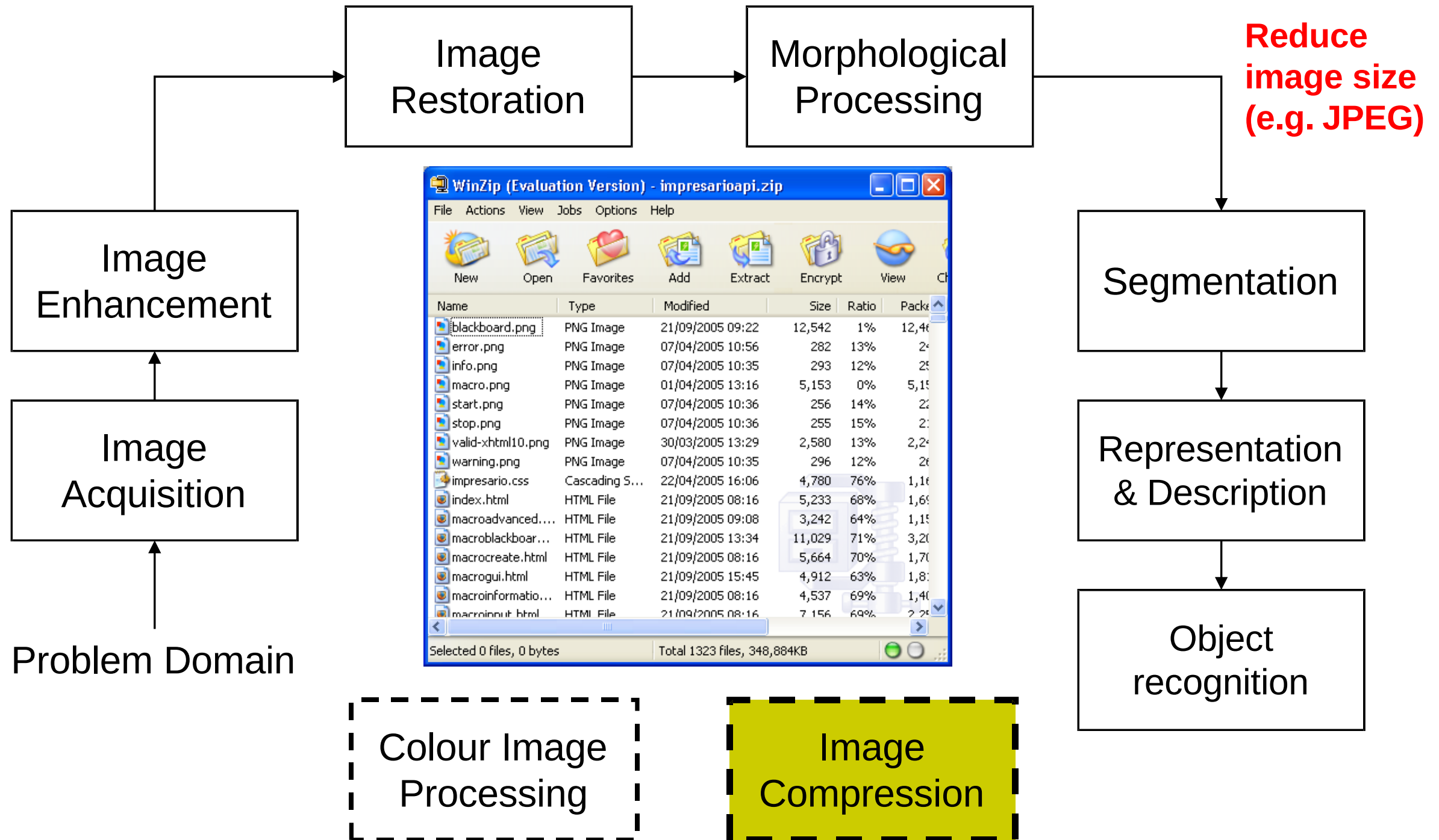
Key Stages in Digital Image Processing: Object Recognition



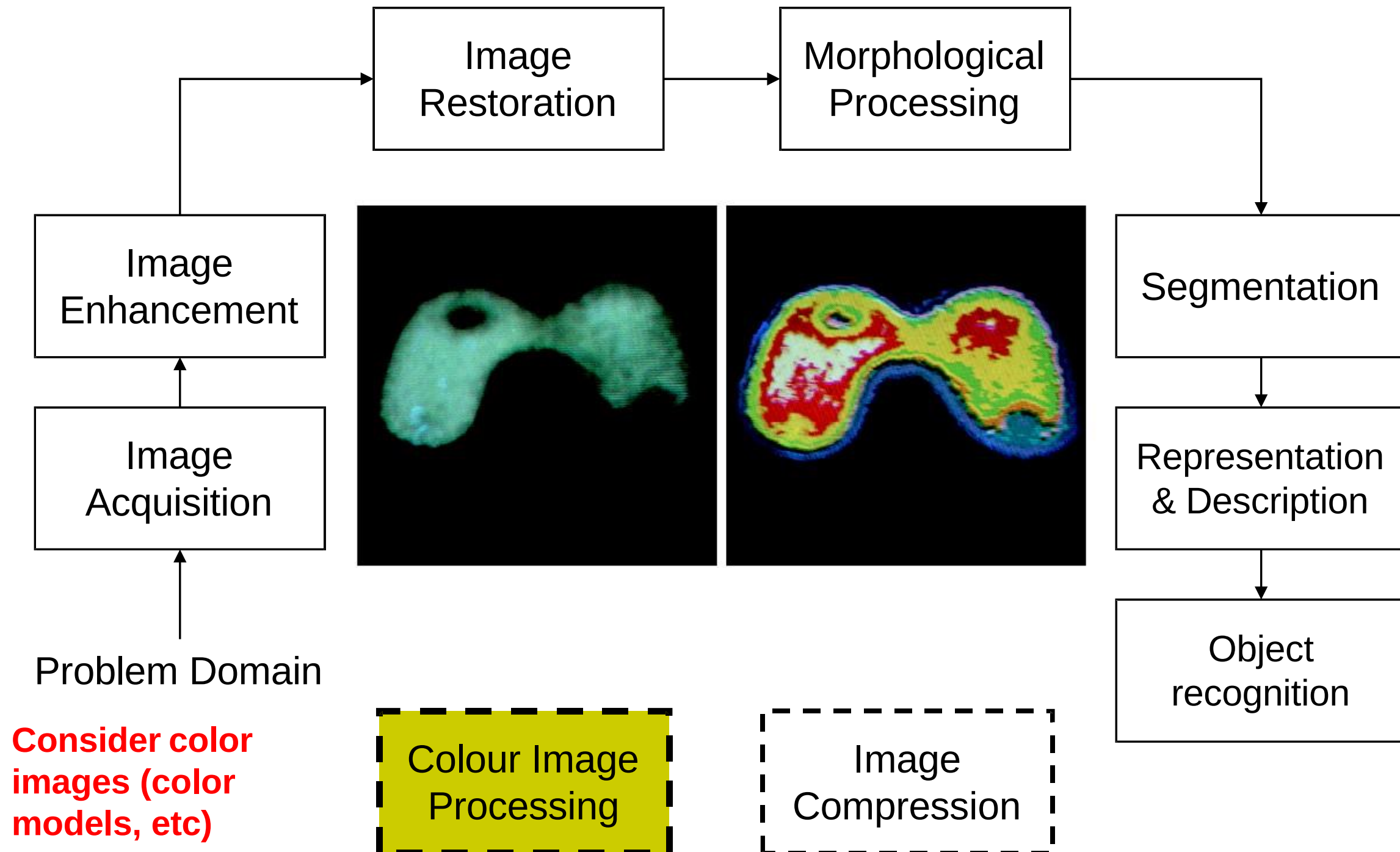
Images taken from Gonzalez & Woods, Digital Image Processing (2002)



Key Stages in Digital Image Processing: Image Compression



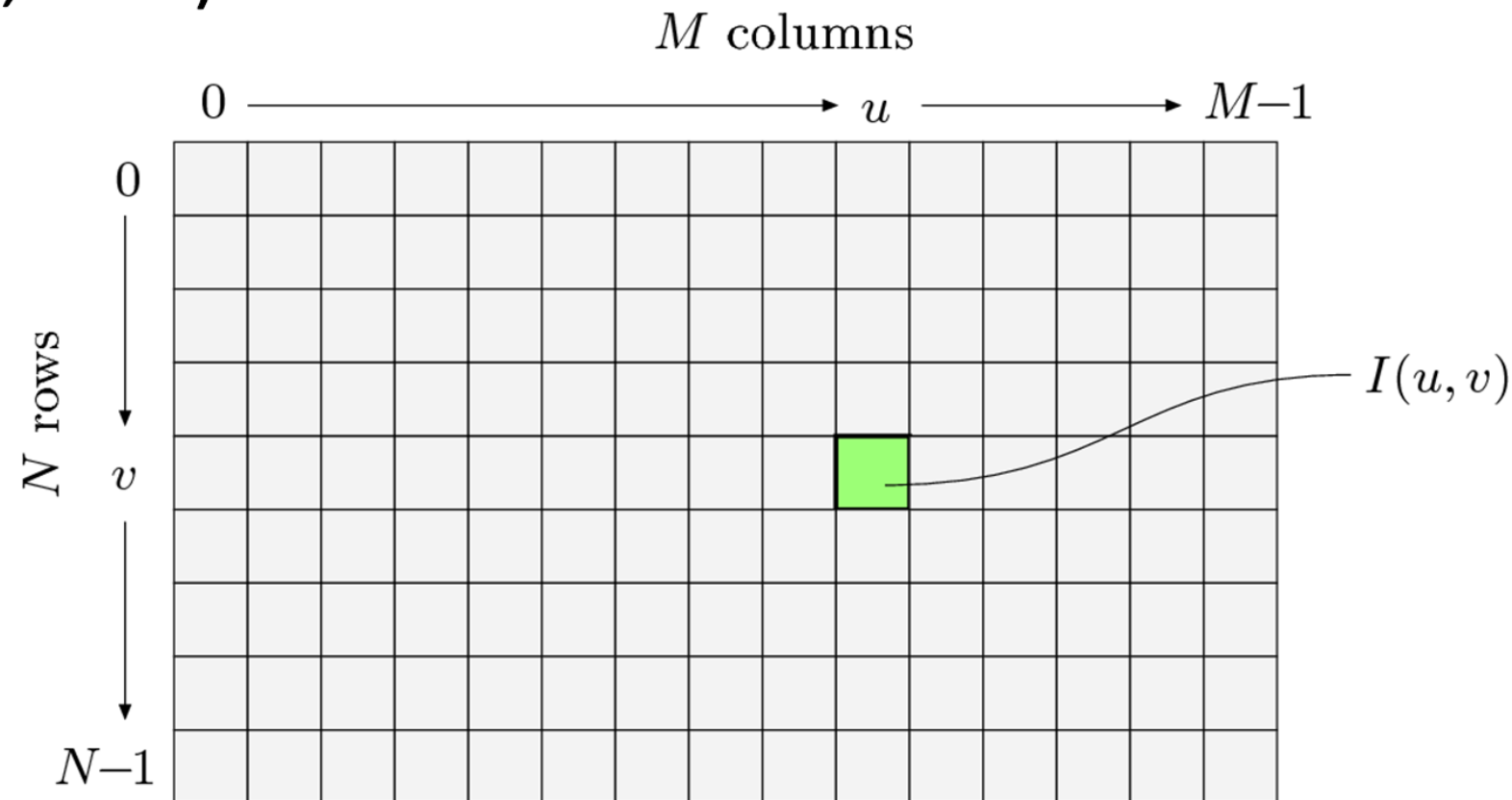
Key Stages in Digital Image Processing: Colour Image Processing



Representing Images



- Image data structure is 2D array of pixel values
- Pixel values are gray levels in range 0-255 or RGB colors
- Array values can be any data type (bit, byte, int, float, double, etc.)



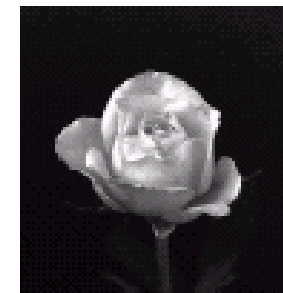


Spatial Resolution

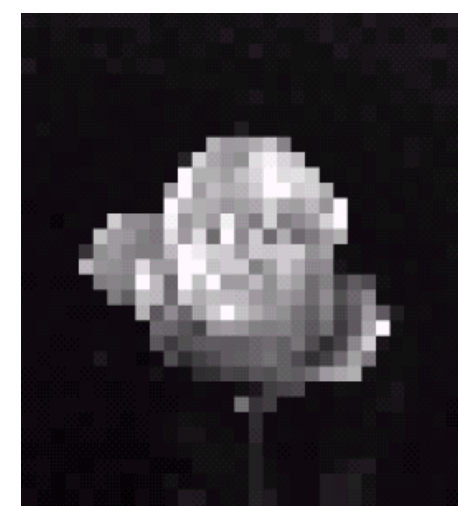
- The **spatial resolution** of an image is determined by how fine/coarse sampling was carried out
- **Spatial resolution:** smallest discernable image detail
 - Vision specialists talk about image resolution
 - Graphic designers talk about *dots per inch* (DPI)



Spatial Resolution



1024





Intensity Level Resolution

- **Intensity level resolution:** number of intensity levels used to represent the image
 - The more intensity levels used, the finer the level of detail discernable in an image
 - Intensity level resolution usually given in terms of number of bits used to store each intensity level

Number of Bits	Number of Intensity Levels	Examples
1	2	0, 1
2	4	00, 01, 10, 11
4	16	0000, 0101, 1111
8	256	00110011, 01010101
16	65,536	1010101010101010

Intensity Level Resolution



256 grey levels (8 bits per pixel)



128 grey levels (7 bpp)



64 grey levels (6 bpp)



32 grey levels (5 bpp)



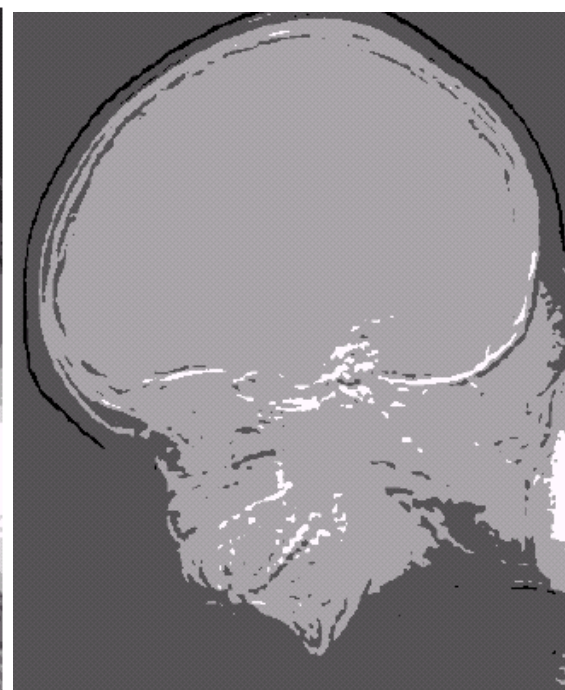
16 grey levels (4 bpp)



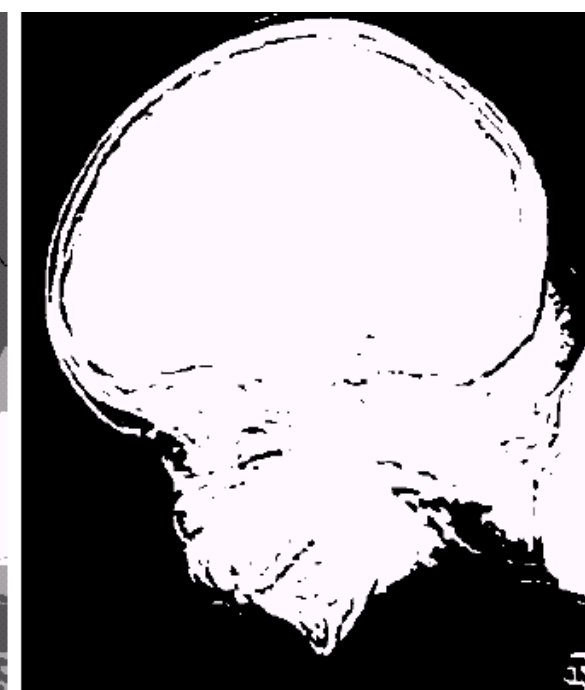
8 grey levels (3 bpp)



4 grey levels (2 bpp)



2 grey levels (1 bpp)



Resolution: How Much Is Enough?



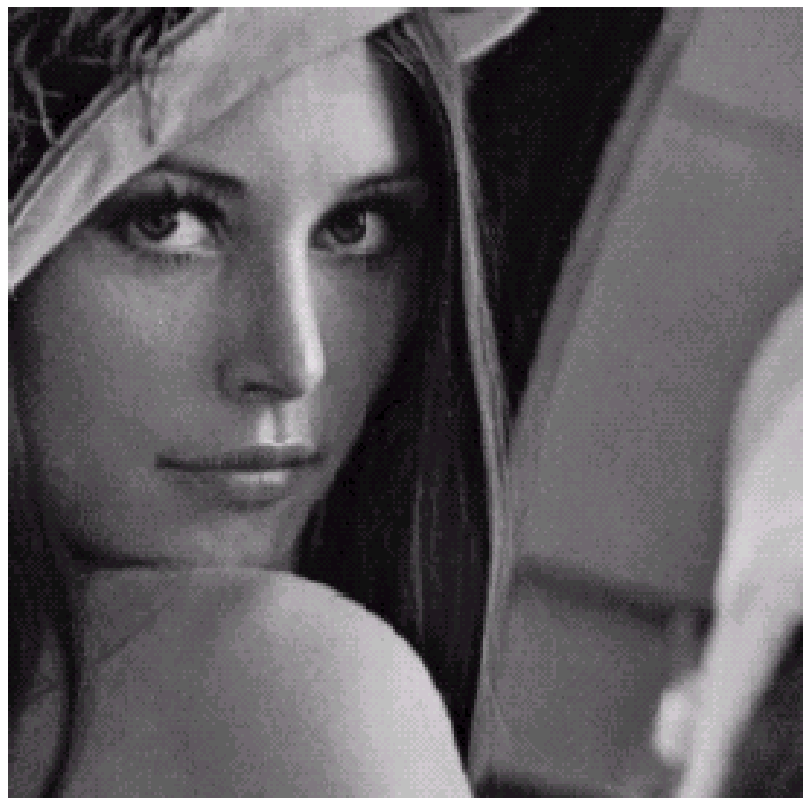
- The big question with resolution is always *how much is enough?*
 - Depends on what is in the image (*details*) and what you would like to do with it (*applications*)
 - Key questions:
 - Does image look aesthetically pleasing?
 - Can you see what you need to see in image?

Resolution: How Much Is Enough?



- **Example:** Picture on right okay for counting number of cars, but not for reading the number plate

Intensity Level Resolution



Low Detail



Medium Detail



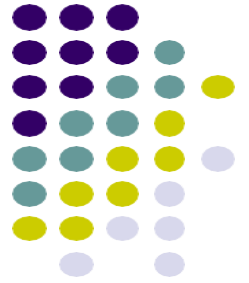
High Detail



Image File Formats

- Hundreds of image file formats. Examples
 - Tagged Image File Format (TIFF)
 - Graphics Interchange Format (GIF)
 - Portable Network Graphics (PNG)
 - JPEG, BMP, Portable Bitmap Format (PBM), etc
- Image pixel values can be
 - **Grayscale:** 0 - 255 range
 - **Binary:** 0 or 1
 - **Color:** RGB colors in 0-255 range (or other color model)
 - **Application specific** (e.g. floating point values in astronomy)

References



- Wilhelm Burger and Mark J. Burge, Digital Image Processing, Springer, 2008
- Gonzales and Woods, Digital Image Processing (3rd edition), Prentice Hall
- Digital Image Processing slides by Brian Mac Namee